



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO THIRU MURUGAN DIESELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

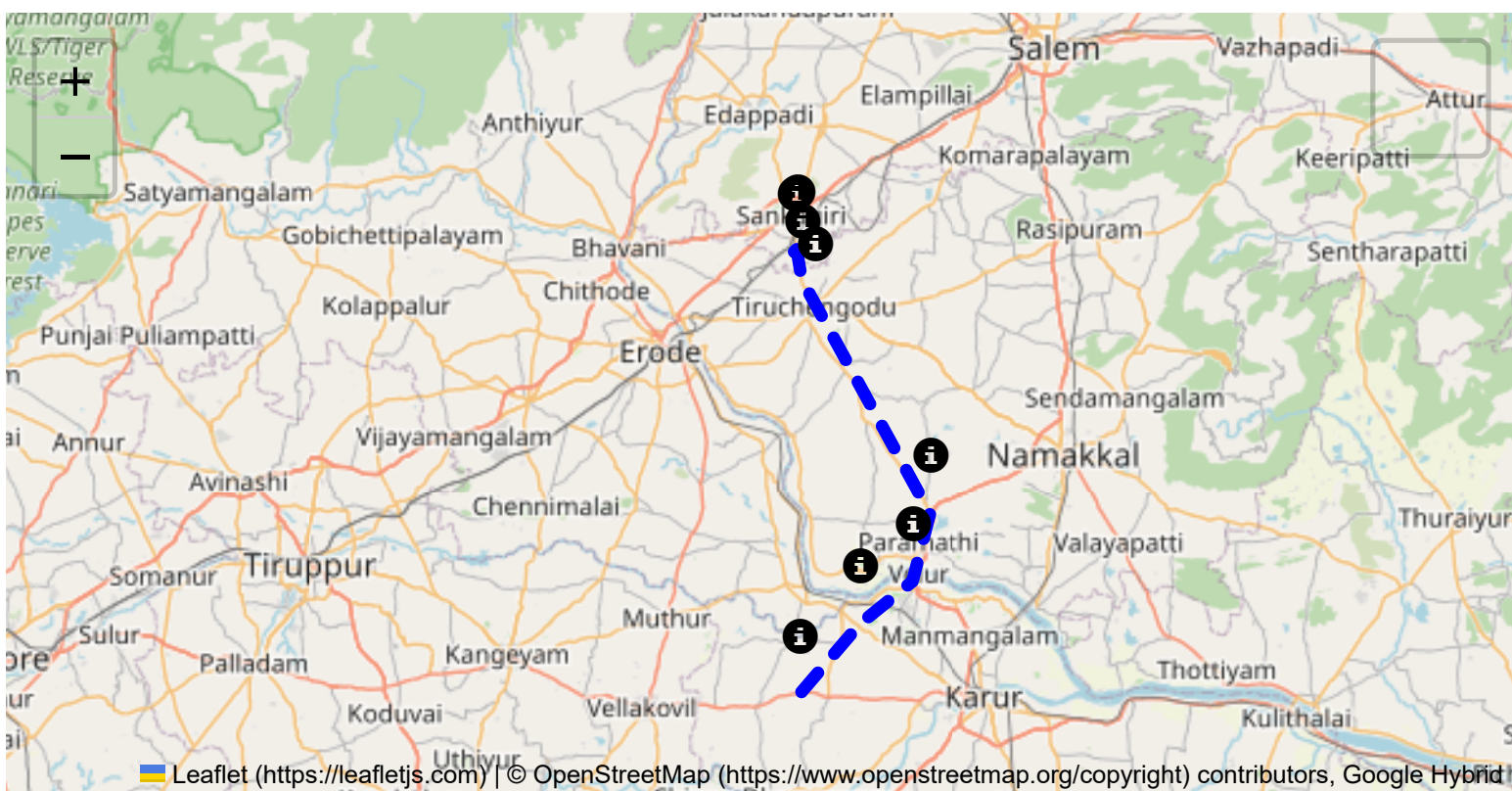
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

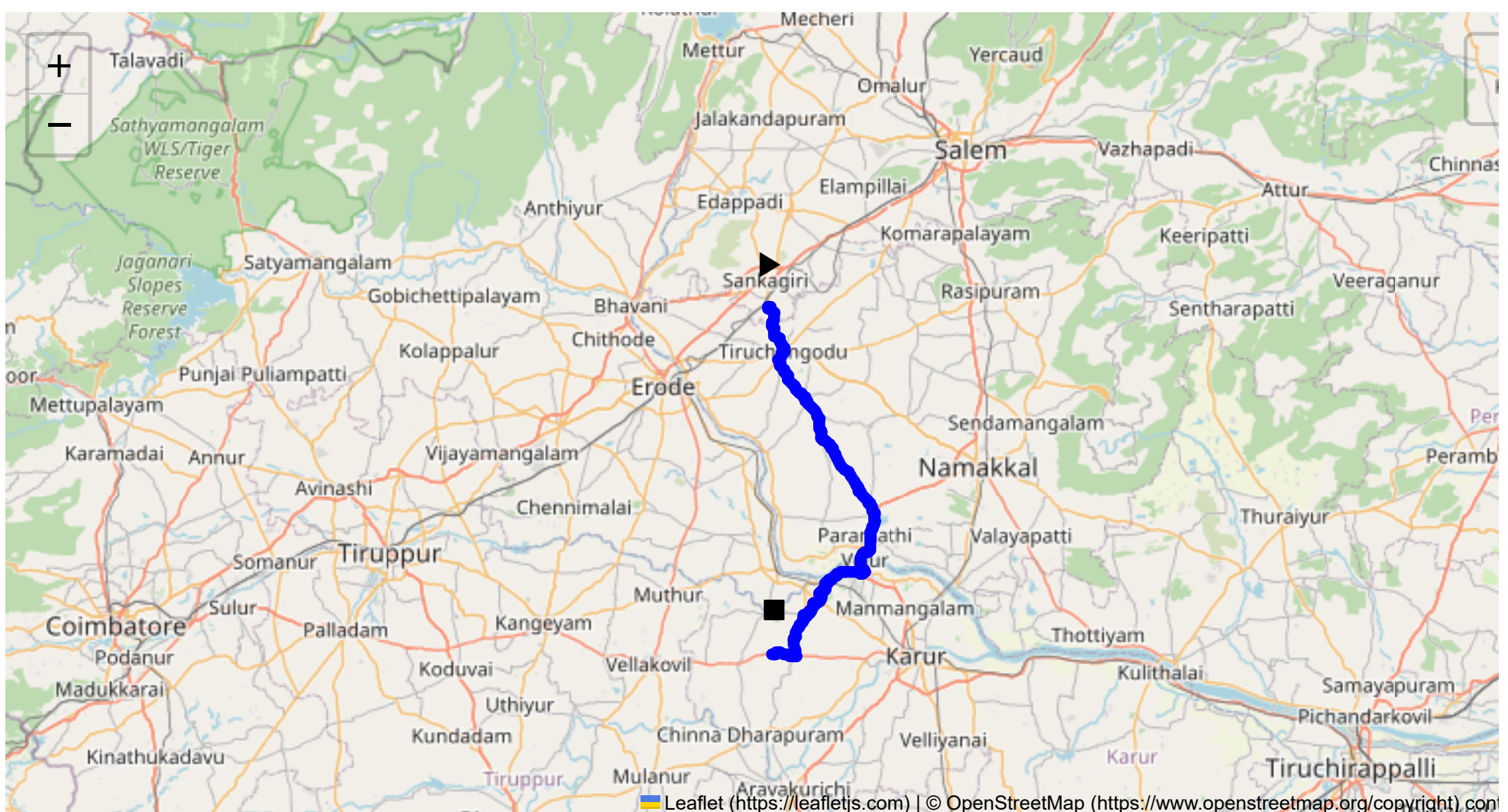
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 71.86 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours
Start: (11.4381, 77.8734)
End: (10.96337421, 77.8805339)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route covers approximately 71.86 kilometers, traveling from Sangagiri through various towns including Kangeyampalayam, Tiruchengode, Paramathi, and Vettamangalam, before reaching the final destination at Nagapattinam - Coimbatore - Gundlupet Hwy. The journey navigates primarily via arterial and state highways, with potential interspersing segments of smaller roads in the towns.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with hot summers and moderate to heavy monsoon rains from October to December. The region may see occasional cyclonic conditions. Monsoon-related hazards such as flooding and reduced visibility due to heavy rainfall are potential risks, particularly in the months of October and November.

3. Analysis of Traffic Patterns

Traffic tends to be heavier during morning (7 AM - 10 AM) and evening (5 PM - 8 PM) peak hours, especially in urban centers like Tiruchengode. Congestion-prone areas are likely around the Tiruchengode Central Bus Stand and intersections with major highways like NH 44.

4. Assessment of Road Quality and Infrastructure

The roads are generally well-maintained, particularly the National Highway portions. However, town roads may vary in quality, with some patches possible in need of maintenance. It's important to be cautious of narrow lanes and speed bumps within towns such as Tiruchengode and Paramathi.

5. Suggestions for Alternative Routes for Emergencies

- If congestion or roadblocks occur, detours using local roads around Tiruchengode or Paramathi could be useful.
- Utilize secondary routes that parallel NH 44 in case of significant highway incidents.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu imposes restrictions on hazardous material transport through densely populated areas and during peak hours to minimize risks. Compliance with specific permits and safety regulations is essential. Frequent checks may be encountered at toll booths and check-posts.

7. Overview of Historical Incidents

There have been sporadic incidents involving overturned heavy vehicles or hazardous material spills primarily due to poor weather conditions or human error. These incidents tend to disrupt traffic for hours and are most common during the monsoon season.

8. Environmental Considerations and Sensitive Areas

- The route traverses agricultural lands and potentially sensitive ecological zones, primarily near water bodies and densely vegetated areas.
- Care should be taken to prevent contamination from spills or emissions.

9. Analysis of Communication Coverage

Mobile network coverage is generally reliable along the route, though there may be brief dead zones when passing through rural patches or dense vegetation at certain segments.

10. Estimated Emergency Response Times

- Urban areas like Tiruchengode: 15-30 minutes
- Rural segments like Vettamangalam: 30-60 minutes
- Proximity to services may increase response times; truck operators should identify nearest hospitals and police stations en route.

11. An Overall Summary of Risk Assessment

The route presents moderate risk levels mainly due to weather and traffic factors. While infrastructure is typically good, heavy rain can significantly increase risk factors. Compliance with local transport regulations is paramount, and operators should be prepared for environmental and traffic-related disruptions. Proper planning, including route monitoring and weather assessment, is advised to minimize potential hazards.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	11.38386, 77.89482	15 KM/Hr
2	Roundabout	High	11.07462, 78.00608	15 KM/Hr
3	Roundabout	High	11.07485, 78.00156	15 KM/Hr
4	U-Turn	High	11.3839218, 77.8950042	10 KM/Hr
5	Turn	Medium	11.43822, 77.87348	30 KM/Hr
6	Turn	High	11.43968, 77.87345	15 KM/Hr
7	Turn	High	11.44022, 77.87546	15 KM/Hr
8	Turn	Medium	11.39671, 77.88878	30 KM/Hr
9	Turn	High	11.37862, 77.89488	15 KM/Hr
10	Turn	High	11.37850, 77.89453	15 KM/Hr
11	Turn	Medium	11.15573, 78.02176	30 KM/Hr
12	Turn	High	11.14545, 78.02208	15 KM/Hr
13	Blind Spot	Blind Spot	11.14480, 78.02291	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	High	11.07466, 78.00842	15 KM/Hr
15	Turn	High	11.05900, 77.95376	15 KM/Hr
16	Turn	Medium	11.04828, 77.94882	30 KM/Hr
17	Turn	High	11.04805, 77.94797	15 KM/Hr
18	Turn	High	11.04668, 77.94762	15 KM/Hr
19	Turn	High	11.04676, 77.94598	15 KM/Hr
20	Turn	High	11.04668, 77.94593	15 KM/Hr
21	Turn	Medium	11.03929, 77.94681	30 KM/Hr
22	Turn	Medium	11.03753, 77.94595	30 KM/Hr
23	Turn	Medium	11.03531, 77.94380	30 KM/Hr
24	Turn	High	11.03330, 77.93776	15 KM/Hr
25	Turn	High	11.01499, 77.92643	15 KM/Hr
26	Blind Spot	Blind Spot	11.01548, 77.92461	10 KM/Hr
27	Turn	Medium	10.95908, 77.91206	30 KM/Hr
28	Turn	High	10.95895, 77.91203	15 KM/Hr
29	Turn	High	10.96313, 77.88043	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
12	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
13	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
15	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
17	hospital	Government Hospital, Velauthapalayam	11.0729521, 78.0023123	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
14	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
16	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38386, 77.89482



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.07462, 78.00608



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.07485, 78.00156



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.3839218, 77.8950042



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44022, 77.87546



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39671, 77.88878



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.15573, 78.02176



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.07466, 78.00842



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.05900, 77.95376



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.04828, 77.94882



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04805, 77.94797



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04668, 77.94762



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.04676, 77.94598



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.04668, 77.94593



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.03929, 77.94681



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.03753, 77.94595



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.03531, 77.94380



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.03330, 77.93776



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.01499, 77.92643



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.01548, 77.92461



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.95908, 77.91206



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.95895, 77.91203



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.96313, 77.88043
